
Database Management Systems
IT354

Project

Deadline: Wednesday 3/12/2025 @ 23:59

[Total Marks for this Assignment is 14]

Student Details:

CRN:

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Instructions:

- You must submit two separate copies (**one Word file and one PDF file**) using the Assignment Template on Blackboard via the allocated folder. These files **must not be in compressed format**.
- It is your responsibility to check and make sure that you have uploaded both the correct files.
- Zero mark will be given if you try to bypass the SafeAssign (e.g. misspell words, remove spaces between words, hide characters, use different character sets, **convert text into image** or languages other than English or any kind of manipulation).
- Email submission will not be accepted.
- You are advised to make your work clear and well-presented. This includes filling your information on the cover page.
- You must use this template, failing which will result in zero mark.
- You **MUST** show all your work, and text must not be converted into an image, unless specified otherwise by the question.
- Late submission will result in ZERO mark.
- The work should be your own, copying from students or other resources will result in ZERO mark.
- Use **Times New Roman** font for all your answers.

*Learning**Outcome(3):*

*Develop a
standard
database using
DBMS.*

Instructions

Imagine you are the database administrator for a large healthcare corporation, and you've been tasked with designing and implementing a new Clinic Management System database. This system addresses real-world needs such as organizing patient records, scheduling appointments, tracking treatments, and handling prescriptions efficiently.

- You must use **MySQL** for database creation and manipulation.
- **All** your answers should be according to the above scenario.
- Support your answer with a **detailed explanation** and **screenshots**.
- You should use the same project document to prepare your answer. **A word file and a pdf file should be provided.**
- Each group can have a **minimum of 2** and a **maximum of 4** students

Learning
Outcome(4):

Analyze
algorithms for
query processing.

Project Description

[14 Marks]

- **Part 1: Database Description and Design**

- 1- Write a brief introduction to a Clinic Management System database, including its purpose and at least five key entities. **(1 Mark)**
- 2- Create an Entity-Relationship Diagram (ERD). Ensure that cardinalities, relationships, and primary keys are clearly represented. State any assumptions that may affect the ER diagram. **(2 Marks)**
- 3- Convert the ER diagram to a class diagram. **(1 Mark)**

- **Part 2: Create and Populate Relations**

- 1- Map the ER diagram to a relational model and create the relations in MySQL **(1 Mark)**
- 2- Enter at least 15 tuples for each relation. **(1 Mark)**
Your solution should include screenshots of your relations with data and the query codes. Otherwise, marks will be deducted

- **Part 3: Indexes, Queries, and Fragmentation Type**

- 1- Write any retrieval query that includes a selection condition. Illustrate how MySQL internally performed this query (e.g., such as parsing and optimization) **(1 Mark)**
- 2- Create an index for the same column used in the previous selection condition. **(1 Mark)**
- 3- Choose a table from your database and apply vertical fragmentation to it. Present the table both before and after the fragmentation process. **(1 Mark)**
- 4- Write a query to retrieve data that includes only one join condition and one selection condition. Then, display the results of the query. **(1 Mark)**
- 5- Show the query statistics and execution plan for the above query. **(1 Mark)**

- **Part 4: Roles, Privileges and Triggers**

- 1- Create 4 roles and assign the following privileges to the roles: **(1 Mark)**
 - Give all privileges to the first role.
 - Give “insert” and “update” privileges to the second role, which can further assign the same privileges to others.
 - Give only select privileges to the third role, which can further assign the same privileges to others.
 - Give “update” privileges to the fourth role.
- 2- Create four accounts and assign the above roles to the created accounts (each account with a different role). **(1 Mark)**
- 3- Create an SQL trigger that verifies a specific condition in a table. **(1 Mark)**

• Part 1: Database Description and Design

- 1- Write a brief introduction to a Clinic Management System database, including its purpose and at least five key entities

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A Clinic Management System (CMS) database is designed to efficiently store, manage, and organize all relevant information for a healthcare clinic. Its main purpose is to facilitate smooth operations like patient registration, appointment scheduling, treatment tracking, and prescription management. By centralizing clinic data, it ensures accuracy, reduces duplication, and improves patient care quality.

key entities:

Entity	Attributes
Patient	(PK)PatientID, Name, Age, Gender, ContactDetails, MedicalHistory
Doctor	(PK)DoctorID, Name, Specialty, ContactDetails, Schedule
Appointment	(PK)AppointmentID, Patient, DoctorID, Date, Time, Duration
Treatment	(PK)TreatmentID, AppointmentID, Diagnosis, Procedure
Prescription	(PK)PrescriptionID, TreatmentID, MedicineID, Dosage, Duration

Medicine	MedicineID, Name, Type, Stock
Prescribed_Medicine	(PK) PrescriptionID, (PK) MedicineID , Dosage, Duration

explanation of the entity:

1. **Patient** – stores patient information (name, age, contact, medical history).
2. **Doctor** – details about doctors (specialty, contact, schedule).
3. **Appointment** – tracks booking details between patients and doctors.
4. **Treatment** – records procedures, diagnoses, and treatments for patients.
5. **Prescription** – keeps track of medicines prescribed to patients.
6. **Medicine** – Stores details about medicines, including name, type, and available stock.

• ERD (Assumptions Affecting the ERD)

These assumptions define the logic used to determine the relationships and cardinalities in the Clinic Management System ERD:

1. Doctor & Appointments

One Doctor can handle many Appointments (1:N), while each Appointment is assigned to only one Doctor.

2. Patient & Appointments

One Patient can book multiple Appointments (1:N), and each Appointment is linked to a single Patient.

3. Appointment & Treatment

Each Appointment results in exactly one Treatment, and the Treatment belongs to that same Appointment. This makes the relationship (1:1), with **AppointmentID** used as a unique key in the Treatment table.

4. Treatment & Prescription

Each Treatment produces one Prescription, and the Prescription belongs to that Treatment.

The relationship is (1:1), with **TreatmentID** used as a unique key in the Prescription table.

5. Medicine & Prescription

The relationship is Many-to-Many (M:N):

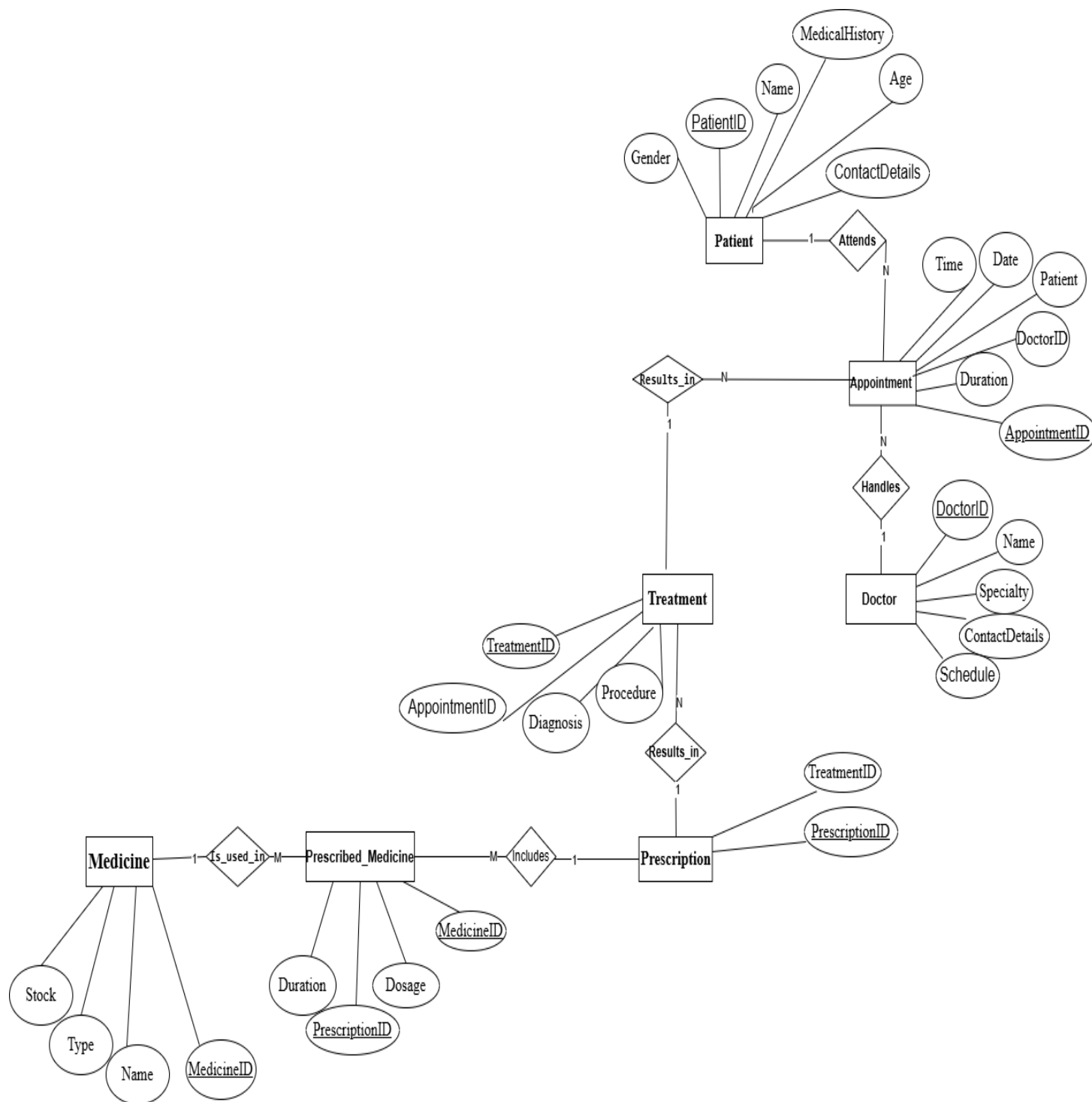
- A Prescription can include multiple Medicines.
- A Medicine can appear in multiple Prescriptions.

This is resolved through a junction entity called **Prescribed_Medicine**.

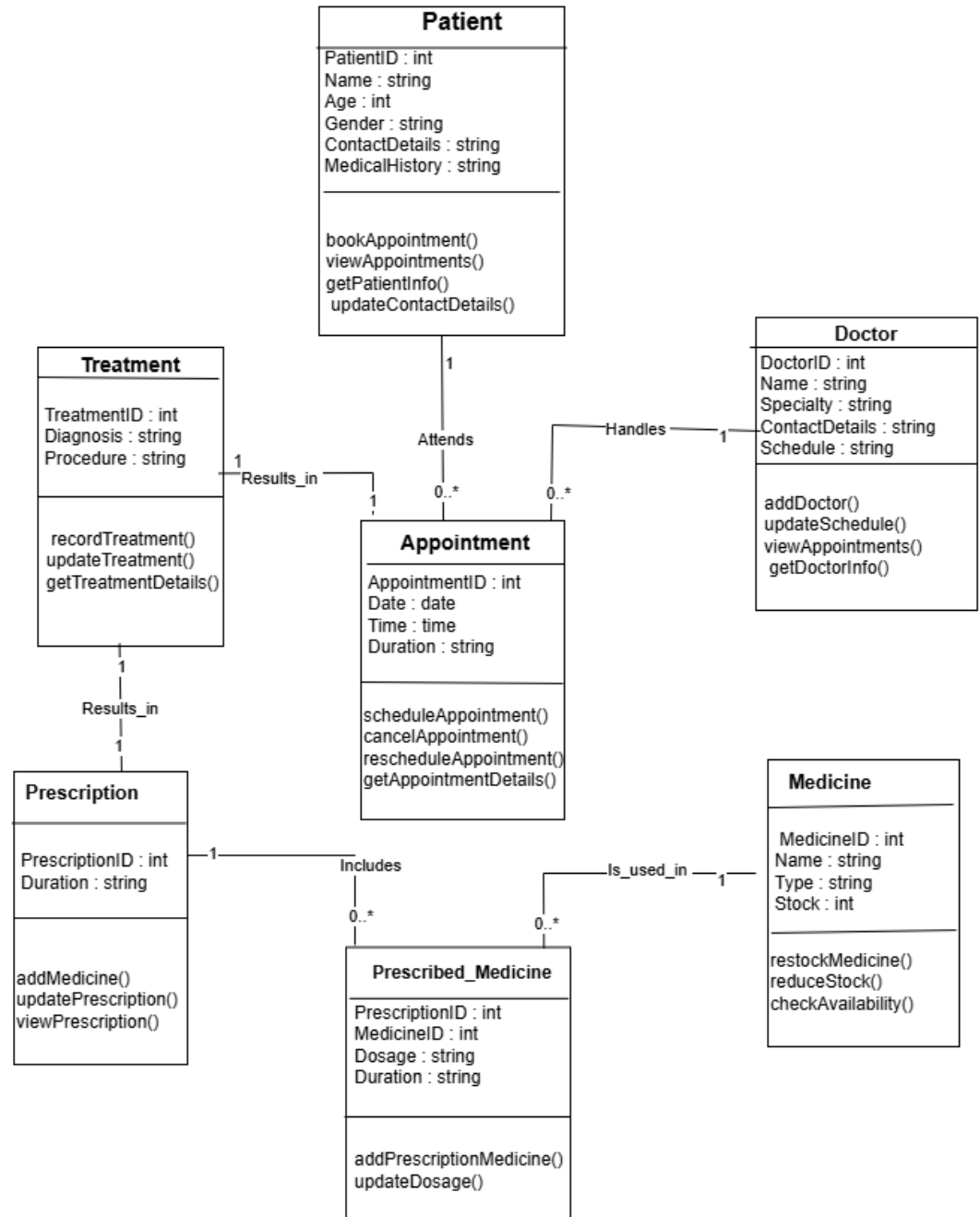
6. Entity Identity

Primary keys (PatientID, DoctorID, AppointmentID, TreatmentID, PrescriptionID, MedicineID) are treated as Surrogate Keys, generated by the system to ensure uniqueness. The junction entity, Prescribed_Medicine, uses a **composite primary key** of (PrescriptionID, MedicineID).

- 2- Create an Entity-Relationship Diagram (ERD). Ensure that cardinalities, relationships, and primary keys are clearly represented. State any assumptions that may affect the ER diagram.



3- Convert the ER diagram to a class diagram.



Part 2

-- DDL: Creating the tables (relations)

-- 1. Patient Table

```
CREATE TABLE Patient (  
    PatientID INT PRIMARY KEY,  
    Name VARCHAR(100) NOT NULL,  
    Age INT,  
    Gender CHAR(1),  
    ContactDetails VARCHAR(255),  
    MedicalHistory TEXT  
);
```

-- 2. Doctor Table

```
CREATE TABLE Doctor (  
    DoctorID INT PRIMARY KEY,  
    Name VARCHAR(100) NOT NULL,  
    Specialty VARCHAR(100),  
    ContactDetails VARCHAR(255),  
    Schedule VARCHAR(100)  
);
```

-- 3. Appointment Table

```
CREATE TABLE Appointment (  
    AppointmentID INT PRIMARY KEY,  
    PatientID INT NOT NULL,  
    DoctorID INT NOT NULL,  
    Date DATE NOT NULL,  
    Time TIME NOT NULL,  
    Duration INT,  
    FOREIGN KEY (PatientID) REFERENCES Patient(PatientID),  
    FOREIGN KEY (DoctorID) REFERENCES Doctor(DoctorID)  
);
```

-- 4. Treatment Table

```
CREATE TABLE Treatment (  

```

```
TreatmentID INT PRIMARY KEY,  
AppointmentID INT NOT NULL UNIQUE,  
Diagnosis VARCHAR(255),  
ProcedureText TEXT,  
FOREIGN KEY (AppointmentID) REFERENCES Appointment(AppointmentID)  
);
```

-- 5. Medicine Table

```
CREATE TABLE Medicine (  
    MedicineID INT PRIMARY KEY,  
    Name VARCHAR(100) NOT NULL,  
    Type VARCHAR(50),  
    Stock INT  
);
```

-- 6. Prescription Table

```
CREATE TABLE Prescription (  
    PrescriptionID INT PRIMARY KEY,  
    TreatmentID INT NOT NULL UNIQUE,  
    FOREIGN KEY (TreatmentID) REFERENCES Treatment(TreatmentID)  
);
```

-- 7. Prescribed_Medicine Table

```
CREATE TABLE Prescribed_Medicine (  
    PrescriptionID INT NOT NULL,  
    MedicineID INT NOT NULL,  
    Dosage VARCHAR(50),  
    Duration VARCHAR(50),  
    PRIMARY KEY (PrescriptionID, MedicineID),  
    FOREIGN KEY (PrescriptionID) REFERENCES Prescription(PrescriptionID),  
    FOREIGN KEY (MedicineID) REFERENCES Medicine(MedicineID)  
);
```

1. Patient Data

```

INSERT INTO Patient (PatientID, Name, Age, Gender, ContactDetails,
MedicalHistory) VALUES
(1001, 'Tasneem Abdulkarim', 28, 'F', '0501234567', 'Fatigue'),
(1002, 'Noor Ahmed', 22, 'F', '0559876543', 'None'),
(1003, 'Yousef Salim', 10, 'M', '0567891234', 'Allergy to Penicillin'),
(1004, 'Amani Alawbathani', 40, 'F', '0500112233', 'Diabetes'),
(1005, 'Faris Majid', 31, 'M', '0554321098', 'Fractured arm (old)'),
(1006, 'Hanadi Qasem', 55, 'F', '0561239876', 'Cardiac History'),
(1007, 'Zain Tarek', 5, 'M', '0502345678', 'Recurring ear infections'),
(1008, 'Leena Omar', 77, 'F', '0558765432', 'Arthritis'),
(1009, 'Saud Badr', 39, 'M', '0560987654', 'Chronic back pain'),
(1010, 'Aisha Naser', 28, 'F', '0503456789', 'Migraines'),
(1011, 'Khalid Musa', 19, 'M', '0557654321', 'Sports injury'),
(1012, 'Rehab Zaid', 49, 'F', '0565432109', 'None'),
(1013, 'Jasim Turki', 60, 'M', '0504567890', 'Type 2 Diabetes'),
(1014, 'Maha Adil', 33, 'F', '0553210987', 'Vitamin D Deficiency'),
(1015, 'Basel Omar', 15, 'M', '0569012345', 'Routine Checkup');

```

Output:

```
SELECT * FROM Patient;
```

Output:

PatientID	Name	Age	Gender	ContactDetails	MedicalHistory
1001	Tasneem Abdulkarim	28	F	0501234567	Fatigue
1002	Noor Ahmed	22	F	0559876543	None
1003	Yousef Salim	10	M	0567891234	Allergy to Penicillin
1004	Amani Alawbathani	40	F	0500112233	Diabetes
1005	Faris Majid	31	M	0554321098	Fractured arm (old)
1006	Hanadi Qasem	55	F	0561239876	Cardiac History
1007	Zain Tarek	5	M	0502345678	Recurring ear infections
1008	Leena Omar	77	F	0558765432	Arthritis
1009	Saud Badr	39	M	0560987654	Chronic back pain
1010	Aisha Naser	28	F	0503456789	Migraines
1011	Khalid Musa	19	M	0557654321	Sports injury
1012	Rehab Zaid	49	F	0565432109	None
1013	Jasim Turki	60	M	0504567890	Type 2 Diabetes
1014	Maha Adil	33	F	0553210987	Vitamin D Deficiency
1015	Basel Omar	15	M	0569012345	Routine Checkup

```
INSERT INTO Doctor (DoctorID, Name, Specialty, ContactDetails, Schedule) VALUES
(10, 'Dr. Sarah Khan', 'Cardiology', 'drsarah@clinic.com', 'Mon-Fri 8-4'),
(11, 'Dr. Ahmed Almansouri', 'Dermatology', 'drahmed@clinic.com', 'Tue-Sat 9-5'),
(12, 'Dr. Fatima Hassan', 'Pediatrics', 'drfatima@clinic.com', 'Mon, Wed, Fri 10-6'),
(13, 'Dr. Khalid Saeed', 'General Practice', 'drkhalid@clinic.com', 'Daily 8-4'),
(14, 'Dr. Mona Fares', 'Orthopedics', 'drmona@clinic.com', 'Mon-Thu 11-7'),
(15, 'Dr. Yousef Sami', 'Neurology', 'dryousef@clinic.com', 'Sun-Wed 7-3'),
(16, 'Dr. Laila Omar', 'Ophthalmology', 'drlaila@clinic.com', 'Tue-Fri 8-4'),
(17, 'Dr. Saad Nasir', 'Internal Medicine', 'drsaad@clinic.com', 'Mon-Sat 12-8'),
(18, 'Dr. Huda Mansour', 'Endocrinology', 'drhuda@clinic.com', 'Wed-Sun 9-5'),
(19, 'Dr. Omar Bakri', 'Physiotherapy', 'dromar@clinic.com', 'Daily 10-6'),
(20, 'Dr. Nora Zaki', 'Cardiology', 'drnora@clinic.com', 'Mon-Fri 1-9'),
(21, 'Dr. Majed Salim', 'Pediatrics', 'drmajed@clinic.com', 'Tue, Thu 7-5'),
(22, 'Dr. Amal Qasim', 'General Practice', 'dramal@clinic.com', 'Daily 9-5'),
(23, 'Dr. Tarek Fawaz', 'Dermatology', 'drtarek@clinic.com', 'Mon, Wed, Fri 11-7'),
(24, 'Dr. Reem Adel', 'Orthopedics', 'drreem@clinic.com', 'Tue-Fri 10-6');
```

Output:

```
SELECT * FROM Doctor;
```

Output:

DoctorID	Name	Specialty	ContactDetails	Schedule
10	Dr. Sarah Khan	Cardiology	drsarah@clinic.com	Mon-Fri 8-4
11	Dr. Ahmed Almansouri	Dermatology	drahmed@clinic.com	Tue-Sat 9-5
12	Dr. Fatima Hassan	Pediatrics	drfatima@clinic.com	Mon, Wed, Fri 10-6
13	Dr. Khalid Saeed	General Practice	drkhalid@clinic.com	Daily 8-4
14	Dr. Mona Fares	Orthopedics	drmona@clinic.com	Mon-Thu 11-7
15	Dr. Yousef Sami	Neurology	dryousef@clinic.com	Sun-Wed 7-3
16	Dr. Laila Omar	Ophthalmology	drlaila@clinic.com	Tue-Fri 8-4
17	Dr. Saad Nasir	Internal Medicine	drsaad@clinic.com	Mon-Sat 12-8
18	Dr. Huda Mansour	Endocrinology	drhuda@clinic.com	Wed-Sun 9-5
19	Dr. Omar Bakri	Physiotherapy	dromar@clinic.com	Daily 10-6
20	Dr. Nora Zaki	Cardiology	drnora@clinic.com	Mon-Fri 1-9
21	Dr. Majed Salim	Pediatrics	drmajed@clinic.com	Tue, Thu 7-5
22	Dr. Amal Qasim	General Practice	dramal@clinic.com	Daily 9-5
23	Dr. Tarek Fawaz	Dermatology	drtarek@clinic.com	Mon, Wed, Fri 11-7
24	Dr. Reem Adel	Orthopedics	drreem@clinic.com	Tue-Fri 10-6

3- Appointment Data

```
INSERT INTO Appointment (AppointmentID, PatientID, DoctorID, Date, Time,
Duration) VALUES
```

```
(1, 1001, 13, '2025-03-01', '09:00:00', 30),
(2, 1002, 13, '2025-03-01', '10:00:00', 20),
(3, 1003, 12, '2025-03-01', '11:00:00', 25),
(4, 1004, 18, '2025-3-02', '13:00:00', 40),
(5, 1005, 14, '2025-03-02', '14:30:00', 30),
(6, 1006, 10, '2025-03-03', '08:30:00', 30),
(7, 1007, 21, '2025-03-03', '10:00:00', 20),
(8, 1008, 17, '2025-03-04', '11:30:00', 30),
(9, 1009, 19, '2025-3-04', '15:00:00', 45),
(10, 1010, 22, '2025-03-05', '09:30:00', 20),
(11, 1011, 14, '2025-03-05', '16:00:00', 30),
(12, 1012, 11, '2025-03-06', '10:30:00', 20),
(13, 1013, 18, '2025-3-06', '14:00:00', 40),
(14, 1014, 22, '2025-03-07', '09:00:00', 20),
(15, 1015, 13, '2025-03-07', '11:00:00', 20);
```

Output:

```
SELECT * FROM Appointment;
```

Output:

AppointmentID	PatientID	DoctorID	Date	Time	Duration
1	1001	13	2025-03-01	09:00:00	30
2	1002	13	2025-03-01	10:00:00	20
3	1003	12	2025-03-01	11:00:00	25
4	1004	18	2025-03-02	13:00:00	40
5	1005	14	2025-03-02	14:30:00	30
6	1006	10	2025-03-03	08:30:00	30
7	1007	21	2025-03-03	10:00:00	20
8	1008	17	2025-03-04	11:30:00	30
9	1009	19	2025-03-04	15:00:00	45
10	1010	22	2025-03-05	09:30:00	20
11	1011	14	2025-03-05	16:00:00	30
12	1012	11	2025-03-06	10:30:00	20
13	1013	18	2025-03-06	14:00:00	40
14	1014	22	2025-03-07	09:00:00	20
15	1015	13	2025-03-07	11:00:00	20

```

INSERT INTO Treatment (TreatmentID, AppointmentID, Diagnosis, ProcedureText)
VALUES
(101, 1, 'General Fatigue', 'Blood work ordered; advice on sleep hygiene.'),
(102, 2, 'Common Cold', 'Symptomatic treatment provided.'),
(103, 3, 'Viral Infection', 'Prescribed medication and rest advised.'),
(104, 4, 'Uncontrolled Diabetes', 'Insulin dosage adjusted; nutritional counseling.'),
(105, 5, 'Shoulder Strain', 'Recommended physiotherapy and anti-inflammatory medication.'),
(106, 6, 'Cardiac Arrhythmia', 'ECG ordered; initial medication prescribed.'),
(107, 7, 'Otitis Media (Ear Infection)', 'Antibiotics prescribed.'),
(108, 8, 'Osteoarthritis flare-up', 'Pain management medication adjusted.'),
(109, 9, 'Sciatica', 'Provided back exercises and heat therapy recommendations.'),
(110, 10, 'Tension Headache', 'Stress management techniques and over-the-counter pain relief advised.'),
(111, 11, 'Ankle Sprain (Grade 1)', 'R.I.C.E. protocol and follow-up in one week.'),
(112, 12, 'Acne Vulgaris', 'Topical retinoid and oral antibiotic prescribed.'),
(113, 13, 'Diabetic Foot Check', 'Routine examination; no immediate concerns.'),
(114, 14, 'Fatigue', 'Blood work ordered; preliminary advice on sleep hygiene.'),
(115, 15, 'Routine Examination', 'Vaccination status checked; patient cleared for school sports.');
```

Output:

```
SELECT * FROM Treatment;
```

Output:

TreatmentID	AppointmentID	Diagnosis	ProcedureText
101	1	General Fatigue	Blood work ordered; advice on sleep hygiene.
102	2	Common Cold	Symptomatic treatment provided.
103	3	Viral Infection	Prescribed medication and rest advised.
104	4	Uncontrolled Diabetes	Insulin dosage adjusted; nutritional counseling.
105	5	Shoulder Strain	Recommended physiotherapy and anti-inflammatory medication.
106	6	Cardiac Arrhythmia	ECG ordered; initial medication prescribed.
107	7	Otitis Media (Ear Infection)	Antibiotics prescribed.
108	8	Osteoarthritis flare-up	Pain management medication adjusted.
109	9	Sciatica	Provided back exercises and heat therapy recommendations.
110	10	Tension Headache	Stress management techniques and over-the-counter pain relief advised.
111	11	Ankle Sprain (Grade 1)	R.I.C.E. protocol and follow-up in one week.
112	12	Acne Vulgaris	Topical retinoid and oral antibiotic prescribed.
113	13	Diabetic Foot Check	Routine examination; no immediate concerns.
114	14	Fatigue	Blood work ordered; preliminary advice on sleep hygiene.
115	15	Routine Examination	Vaccination status checked; patient cleared for school sports.

```
INSERT INTO Medicine (MedicineID, Name, Type, Stock) VALUES
(201, 'Amlodipine', 'Antihypertensive', 500),
(202, 'Paracetamol', 'Analgesic', 1200),
(203, 'Amoxicillin', 'Antibiotic', 850),
(204, 'Metformin', 'Antidiabetic', 1500),
(205, 'Ibuprofen', 'NSAID', 900),
(206, 'Beta Blocker X', 'Cardiovascular', 450),
(207, 'Ciprofloxacin', 'Antibiotic', 600),
(208, 'Diclofenac Gel', 'Topical NSAID', 700),
(209, 'Muscle Relaxant A', 'Muscle Relaxant', 350),
(210, 'Vitamin D3', 'Supplement', 1100),
(211, 'Topical Retinoid', 'Dermatological', 400),
(212, 'Doxycycline', 'Oral Antibiotic', 550),
(213, 'Folic Acid', 'Supplement', 950),
(214, 'Cough Syrup B', 'Cough Suppressant', 800),
(215, 'Zolpidem', 'Sleeping Aid', 250);
```

Output:

```
SELECT * FROM Medicine;
```

Output:

MedicineID	Name	Type	Stock
201	Amlodipine	Antihypertensive	500
202	Paracetamol	Analgesic	1200
203	Amoxicillin	Antibiotic	850
204	Metformin	Antidiabetic	1500
205	Ibuprofen	NSAID	900
206	Beta Blocker X	Cardiovascular	450
207	Ciprofloxacin	Antibiotic	600
208	Diclofenac Gel	Topical NSAID	700
209	Muscle Relaxant A	Muscle Relaxant	350
210	Vitamin D3	Supplement	1100
211	Topical Retinoid	Dermatological	400
212	Doxycycline	Oral Antibiotic	550
213	Folic Acid	Supplement	950
214	Cough Syrup B	Cough Suppressant	800
215	Zolpidem	Sleeping Aid	250


```
INSERT INTO Prescription (PrescriptionID, TreatmentID) VALUES
(501, 101),
(502, 102),
(503, 103),
(504, 104),
(505, 105),
(506, 106),
(507, 107),
(508, 108),
(509, 109),
(510, 110),
(511, 111),
(512, 112),
(513, 113),
(514, 114),
(515, 115);
```

Output:

```
SELECT * FROM Prescription
```

Output:

PrescriptionID	TreatmentID
501	101
502	102
503	103
504	104
505	105
506	106
507	107
508	108
509	109
510	110
511	111
512	112
513	113
514	114
515	115

ta

```

INSERT INTO Prescribed_Medicine (PrescriptionID, MedicineID, Dosage, Duration)
VALUES
(501, 202, '500mg as needed', '5 days'),
(502, 202, '500mg as needed', '5 days'),
(503, 202, '500mg as needed', '5 days'),
(504, 204, '500mg twice daily', 'Indefinite'),
(504, 210, '1000 IU daily', '60 days'),
(505, 205, '400mg three times daily', '7 days'),
(506, 206, '25mg daily', 'Indefinite'),
(507, 203, '250mg twice daily', '10 days'),
(508, 208, 'Apply twice daily', '14 days'),
(509, 209, '1 tablet at night', '7 days'),
(510, 202, '500mg as needed', '3 days'),
(511, 205, '400mg twice daily', '5 days'),
(512, 211, 'Apply once daily', '90 days'),
(512, 212, '100mg daily', '30 days'),
(514, 210, '5000 IU weekly', '60 days');

```

Output:

```
SELECT * FROM Prescribed_Medicine;
```

Output:

PrescriptionID	MedicineID	Dosage	Duration
501	202	500mg as needed	5 days
502	202	500mg as needed	5 days
503	202	500mg as needed	5 days
504	204	500mg twice daily	Indefinite
504	210	1000 IU daily	60 days
505	205	400mg three times daily	7 days
506	206	25mg daily	Indefinite
507	203	250mg twice daily	10 days
508	208	Apply twice daily	14 days
509	209	1 tablet at night	7 days
510	202	500mg as needed	3 days
511	205	400mg twice daily	5 days
512	211	Apply once daily	90 days
512	212	100mg daily	30 days
514	210	5000 IU weekly	60 days

- **Part 3: Indexes, Queries, and Fragmentation Type**

- 1- Write any retrieval query that includes a selection condition. Illustrate how MySQL internally performed this query (e.g., such as parsing and optimization) (1 Mark)

Answer

SELECT

AppointmentID, Date, Duration, PatientID, DoctorID

FROM

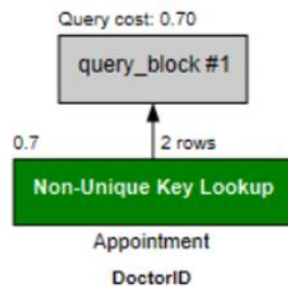
Appointment

WHERE

DoctorID = 22;

How the query was executed is called Execution plan. Here we see the execution plan

is a simple full table scan (brute force) on ALL records with the primary key.



- 2- Create an index for the same column used in the previous selection condition.
(1 Mark)

Answer

**CREATE INDEX index_doctor_id
ON Appointment (DoctorID);**

- 3- Choose a table from your database and apply vertical fragmentation to it.
Present the table both before and after the fragmentation process. (1 Mark)

Answer**Table Before Fragmentation**

Let's take the **Doctor** table as an example:

Doctor(DoctorID, Name, Specialty, ContactDetails, ClinicID)

This table contains all information related to doctors, including their names, specialties, contact details, and the clinic where each doctor works.

Apply Vertical Fragmentation

Vertical fragmentation involves dividing a table **by columns rather than rows**, so each fragment holds a subset of the attributes but includes **all tuples (rows)**.

We can divide the table into two smaller, more specific fragments:

Fragment 1 – Doctor_Basic_Info

This fragment stores the doctor's identification and general information.

```
CREATE TABLE Doctor_Basic_Info AS
```

```
SELECT
```

```
    DoctorID,
```

```
    Name,
```

```
    Specialty
```

```
FROM
```

```
    Doctor;
```

Fragment 2 – Doctor_Contact_Clinic

This fragment contains contact information and clinic details.

```
CREATE TABLE Doctor_Contact_Clinic AS
```

```
SELECT
    DoctorID,
    ContactDetails,
    ClinicID
FROM
    Doctor;
```

Result After Fragmentation

After applying vertical fragmentation, we get two new tables:

- **Doctor_Basic_Info(DoctorID, Name, Specialty)**
- **Doctor_Contact_Clinic(DoctorID, ContactDetails, ClinicID)**

Both fragments include the **DoctorID** attribute, which serves as the **common key** that allows the reconstruction of the original table whenever needed.

Reconstruction of the Original Table

The original table can be re-created using a simple join operation on the common attribute (DoctorID):

```
SELECT
    b.DoctorID,
    b.Name,
    b.Specialty,
    c.ContactDetails,
    c.ClinicID
FROM
    Doctor_Basic_Info b
```

JOIN

Doctor_Contact_Clinic c

ON

b.DoctorID = c.DoctorID;

4. Write a query to retrieve data that includes only one join condition and one selection condition. Then, display the results of the query. (1 Mark)

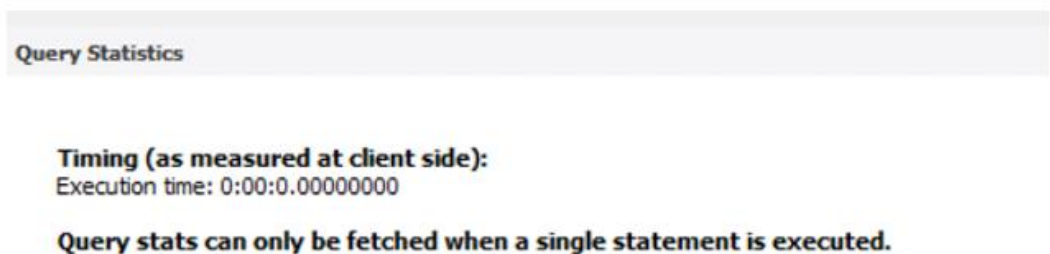
Answer

```
SELECT
a.AppointmentID,
d.Name AS DoctorName
a.Duration
FROM
Appointment a
JOIN
Doctor d ON a.DoctorID = d.DoctorID
WHERE
a.Duration > 35;
```

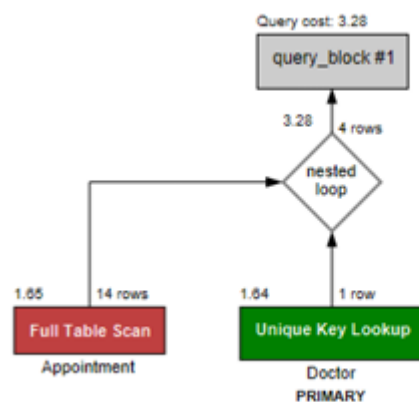
Result Grid		Filter Rows:	Export: 	Wrap Cell Conter
AppointmentID	Date	Duration	DoctorName	
4	2025-3-02 13:00:00	40	Amani Alawbathani	
9	2025-3-04 15:00:00	45	Saud Badr	
13	2025-3-06 14:00:00	40	Jasim Turki	

- 5 - Show the query statistics and execution plan for the above query. (1 Mark)

Query statistics



Execution plan



N

• Part 4: Roles, Privileges and Triggers

1- Create 4 roles and assign the following privileges to the roles: **(1 Mark)**

- Give all privileges to the first role.
- Give “insert” and “update” privileges to the second role, which can further assign the same privileges to others.
- Give only select privileges to the third role, which can further assign the same privileges to others.
- Give “update” privileges to the fourth role.

Answer

```
CREATE ROLE admin_role;  
CREATE ROLE data_entry_role;  
CREATE ROLE auditor_role;  
CREATE ROLE schedule_update_role;
```

```
GRANT ALL PRIVILEGES ON clinic_db.*  
TO admin_role;
```

```
GRANT INSERT, UPDATE ON clinic_db.*  
TO data_entry_role WITH GRANT OPTION;
```

```
GRANT SELECT ON clinic_db.*
```

```
TO auditor_role WITH GRANT OPTION;
```

```
GRANT UPDATE ON clinic_db.Appointment TO schedule_update_role;
```

```
GRANT UPDATE ON clinic_db.Doctor TO schedule_update_role;
```

- 2- Create four accounts and assign the above roles to the created accounts (each account with a different role). **(1 Mark)**

Answer

```
CREATE USER 'dr_super'@'localhost' IDENTIFIED BY 'Pass1234' ;
```

```
CREATE USER 'clerk_data'@'localhost' IDENTIFIED BY 'ClerkPass' ;
```

```
CREATE USER 'auditor_staff'@'localhost' IDENTIFIED BY 'AuditPass' ;
```

```
CREATE USER 'scheduler'@'localhost' IDENTIFIED BY 'SchedPass' ;
```

```
GRANT 'admin_role' TO 'dr_super'@'localhost' ;
```

```
GRANT 'data_entry_role' TO 'clerk_data'@'localhost' ;
```

```
GRANT 'auditor_role' TO 'auditor_staff'@'localhost' ;
```

```
GRANT 'schedule_update_role' TO 'scheduler'@'localhost' ;
```

3- Create SQL trigger that verifies a specific condition in a table. (1 Mark)

Answer

```
CREATE TRIGGER prevent_negative_stock
BEFORE UPDATE ON Medicine
FOR EACH ROW
BEGIN
    IF NEW.Stock < 0 THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'ERROR: Medicine stock cannot be less than zero.' ;
    END IF;
END;
```